PROJECT PROGRESS REPORT

(Progress Report no. 4)

1. Student Information

Student Name	PATEL MAN VISHNUBHAI.
Enrollment Number	25IC03CA053
Program/Branch	Master Of Computer Applications
Semester	3
Contact Number	8849948963
Email ID	25ic03ca053@ppsu.ac.in

2. Project Details

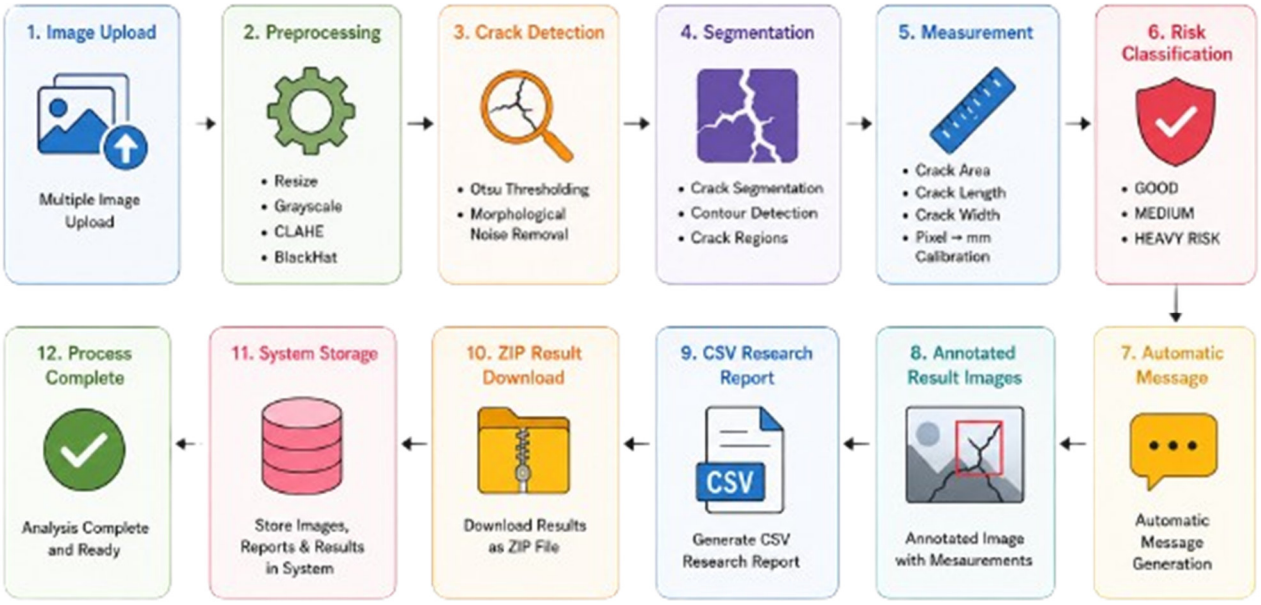
Type	Research Project
Department/Domain	MCA CA(DS/ML) / AI & ML
Mentor Name	Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Work Progress Summary

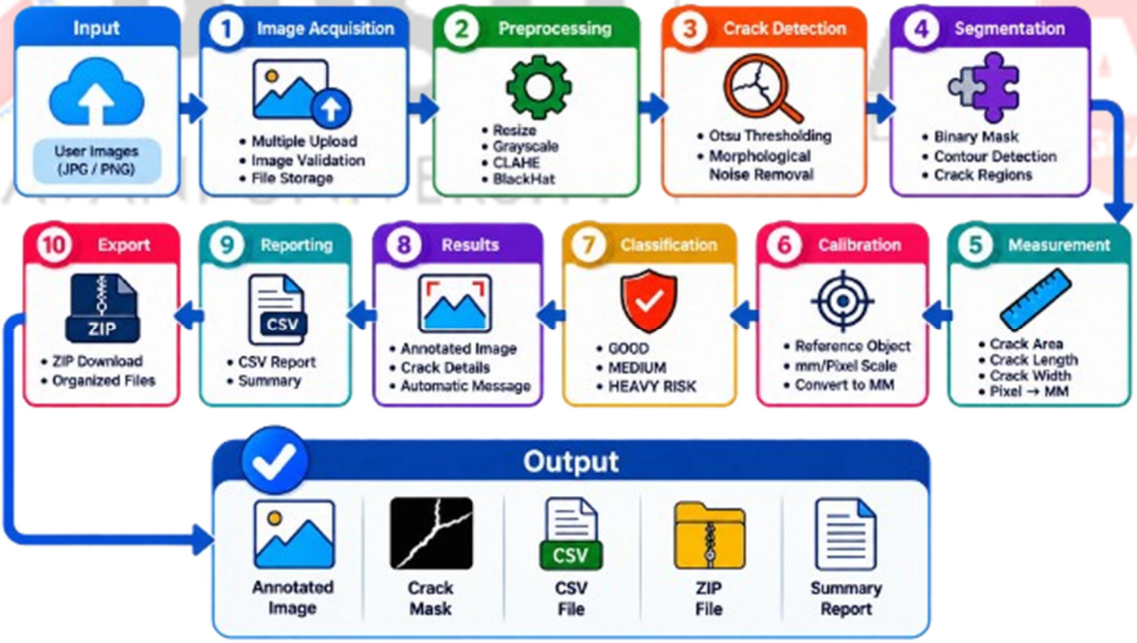
a) Project Title:

- OpenCV - Based System for Crack Identification and Width Estimation

b) Initial Module Implementation Process:



c) Overall Coding Architecture:



d) Code:

a. Library Installation:

Code:

```
!pip -q install opencv-python scikit-image pandas matplotlib

# Check for Working for Install Libraries

print("Install Lbraries complete to Install..")
```

Output:

```
Lbraries Install Work complete..
```

b. Import Libraries:

Code:

```
import cv2
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from google.colab import files
from skimage.morphology import skeletonize

import os
import math
import shutil
import zipfile
from datetime import datetime

from IPython.display import display
```

```
#  Check For Libraries work or not

print("All Libraries Imported Sucessfully..")
```

Output:

```
All Libraries Imported Sucessfully.
```

c. Project Configuration:

Code:

```
Project_name = "OpenCV-Based System For Crack Identification and Width Estimation"

print(Project_name) # Use for Chack in Project name title
```

Output:

```
OpenCV-Based System For Crack Identification and Width Estimation
```

4. Tasks Accomplished

- Implemented Install All Libraries.
- Studied the fundamentals of OpenCV and digital image processing.
- Implemented multiple crack image upload functionality.

5. Tasks in Progress / Next Plan

Task In Progress:

- Output directories
- Optional MM Calibration
- Risk Classification Thresholds
- Upload Multiple Images

Next Plan:

- Crack Length Function
- Risk Classification Function
- Image Analysis Function
- Process All Uploaded Images

6. Learnings / Skills Acquired

- Learned OpenCV and digital image processing fundamentals.
- Learned image preprocessing and enhancement techniques.
- Gained practical knowledge of Grayscale, CLAHE, and BlackHat transformation.
- Learned Otsu Thresholding for crack detection.
- Learned morphological operations for noise removal.
- Learned contour detection and crack segmentation.
- Learned techniques for crack area and length estimation.
- Developed understanding of crack-width estimation and pixel-to-mm calibration.
- Learned risk classification based on crack measurements.
- Gained experience in annotated image/result visualization.
- Learned CSV-based research data generation and result management.
- Improved Python programming and modular coding skills.
- Developed skills in debugging, testing, and validating image-processing modules.

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References:

- **Dorafshan, S., Thomas, R. J., & Maguire, M. (2018).** SDNET2018: An Annotated Image Dataset for Non-Contact Concrete Crack Detection Using Deep Convolutional Neural Networks. Data in Brief, 21, 1664–1668.
- **Zhang, L., Yang, F., Zhang, Y. D., & Zhu, Y. J. (2016).** Road Crack Detection Using Deep Convolutional Neural Network. IEEE International Conference on Image Processing (ICIP).
- **Liu, Y., Yao, J., Lu, X., Xie, R., & Li, L. (2019).** DeepCrack: Learning Hierarchical Convolutional Features for Crack Detection. IEEE Transactions on Image Processing.
- **Dung, C. V., & Anh, L. D. (2019).** Computer Vision-Based Concrete Crack Detection Using U-Net Fully Convolutional Networks. Automation in Construction, 104, 102–113.

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks Obtained (Out of 20)
Problem Definition	L4–Analyze	Clearly defined, justified real-world problem	Defined but lacks depth	Vague problem	No clear problem	
Project Title Relevance	L2–Understand	Highly precise and aligned	Mostly relevant	Partially relevant	Misleading	
Scope of Work	L5–Evaluate	Well-defined scope, deliverables, constraints	Defined with minor gaps	Broad/unclear	No scope	
Related Study	L4–Analyze	3–5 sources analyzed critically	Sources listed	Minimal references	No study	
Clarity & Presentation	L3–Apply	Structured, professional	Minor issues	Some confusion	Poor	
Total Marks						

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Evaluation

Student Signature	Mentor (Company) Signature

Mentor (Institute) Remarks

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